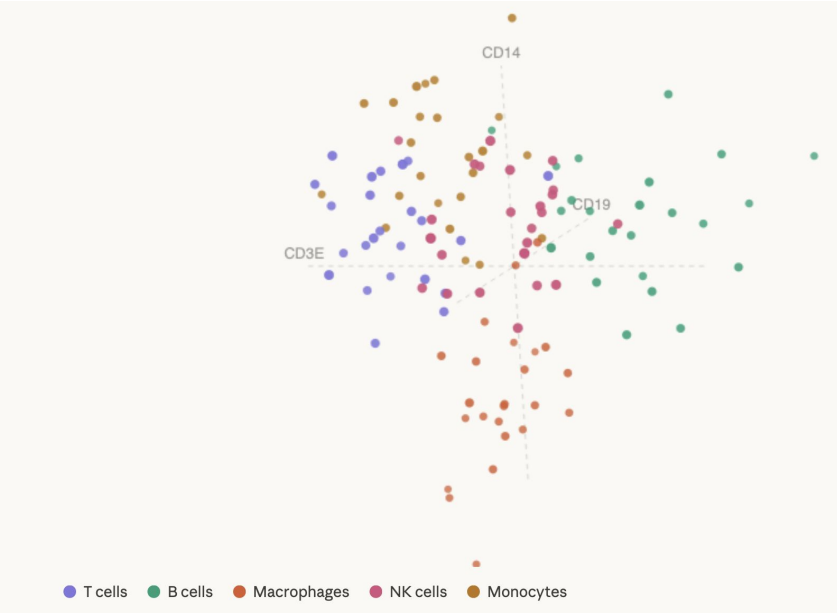
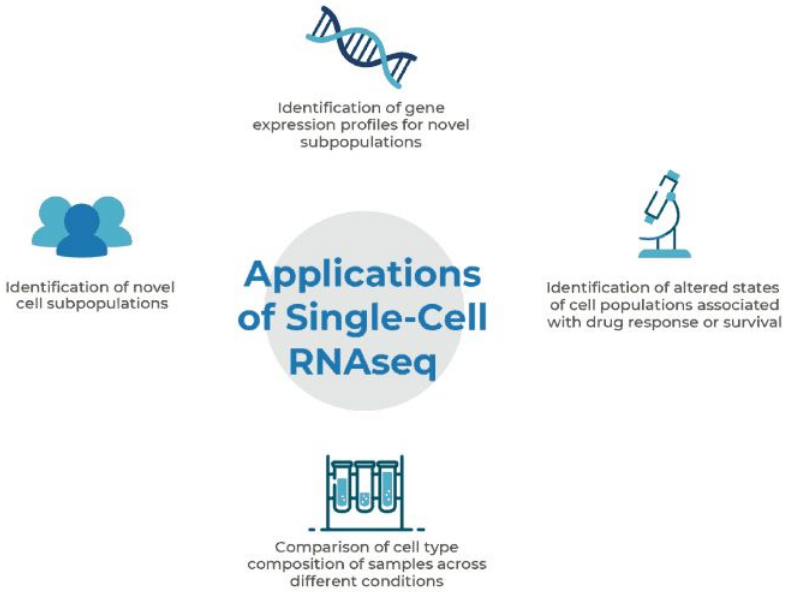
Exploring Gene Expression with Machine Learning

Rika Chan, ERC, April 07, 2026

What is machine learning?

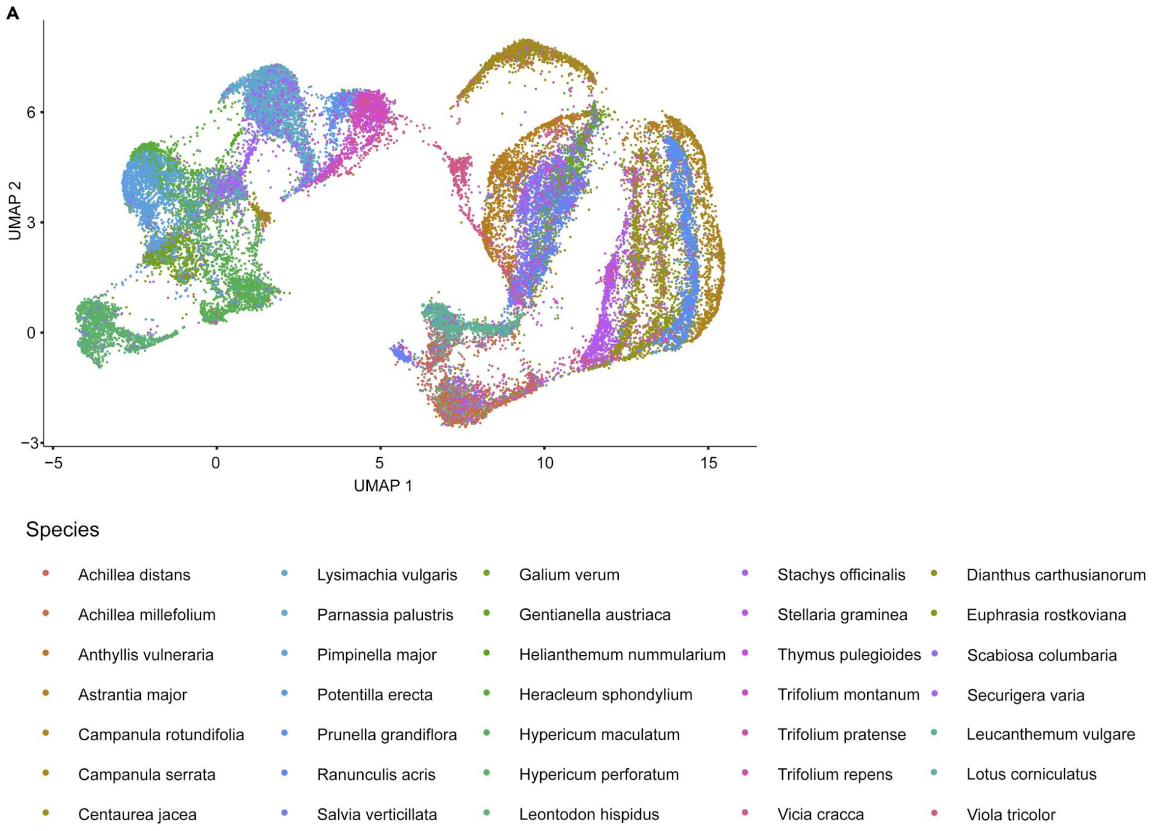
Applications of single-cell research





Fios Genomics

Example plot



(A) 2D UMAP embedding of images from 35 pollen species

Stolarek I, et al. 2022

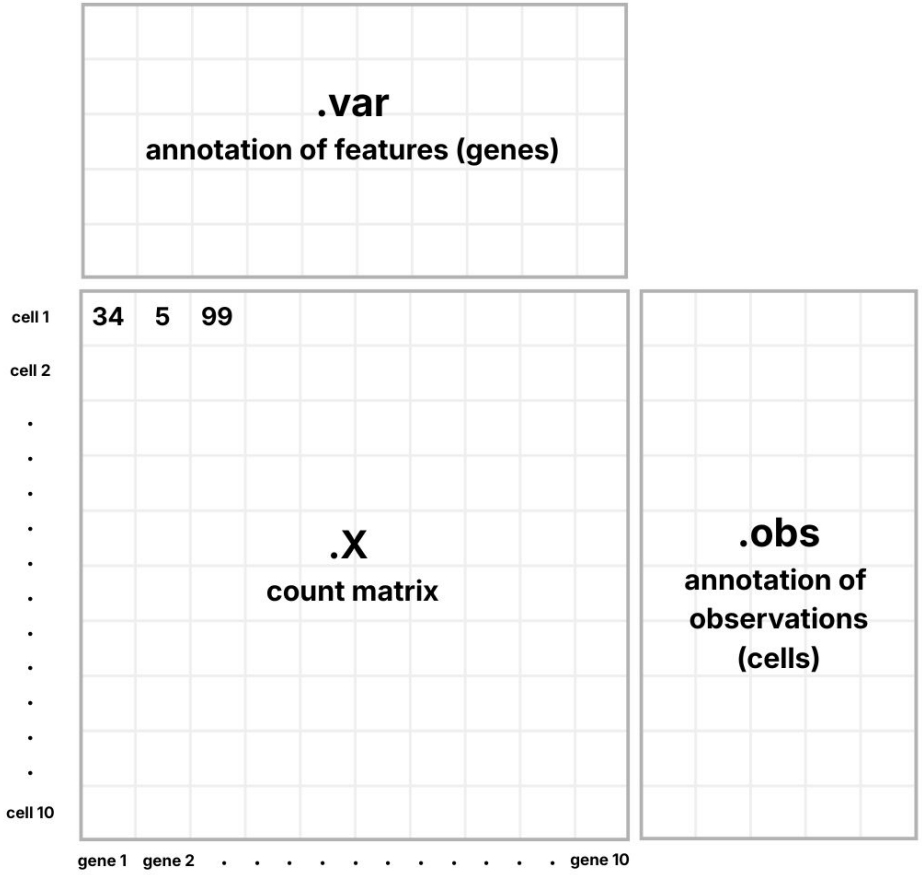
Adata structure

Adata = AnnData

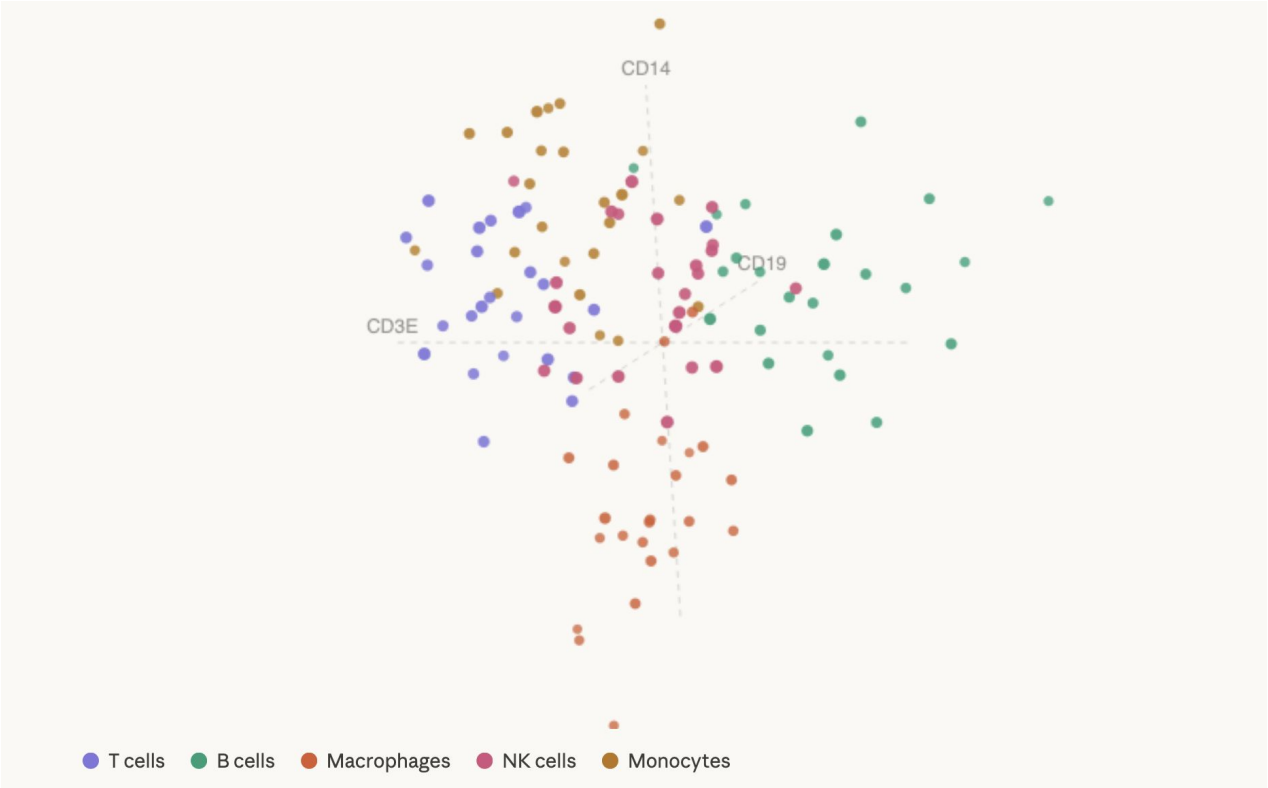
.X = shows # of genes per cell

.obs = cell info

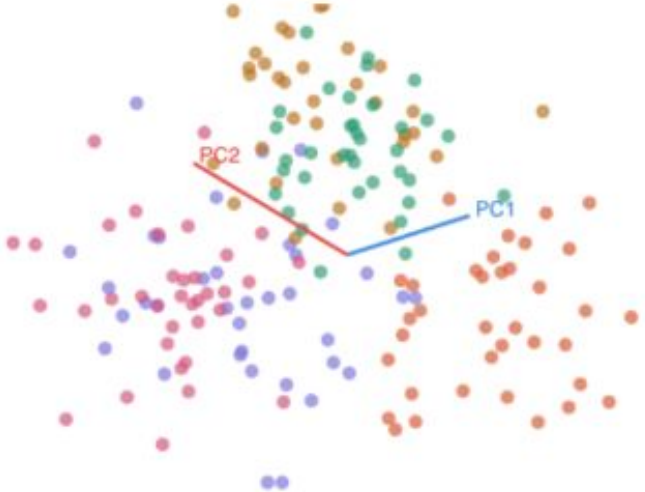
.var = gene info

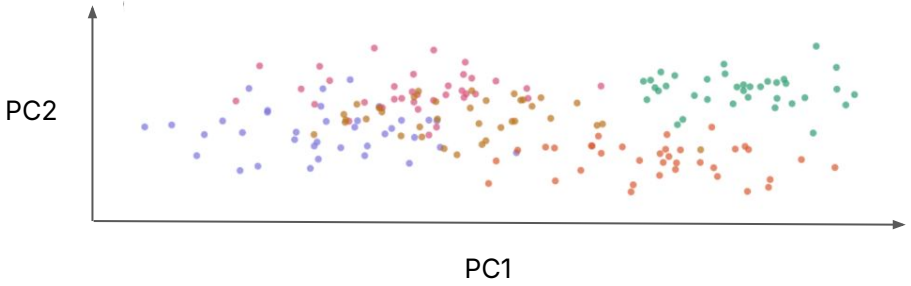


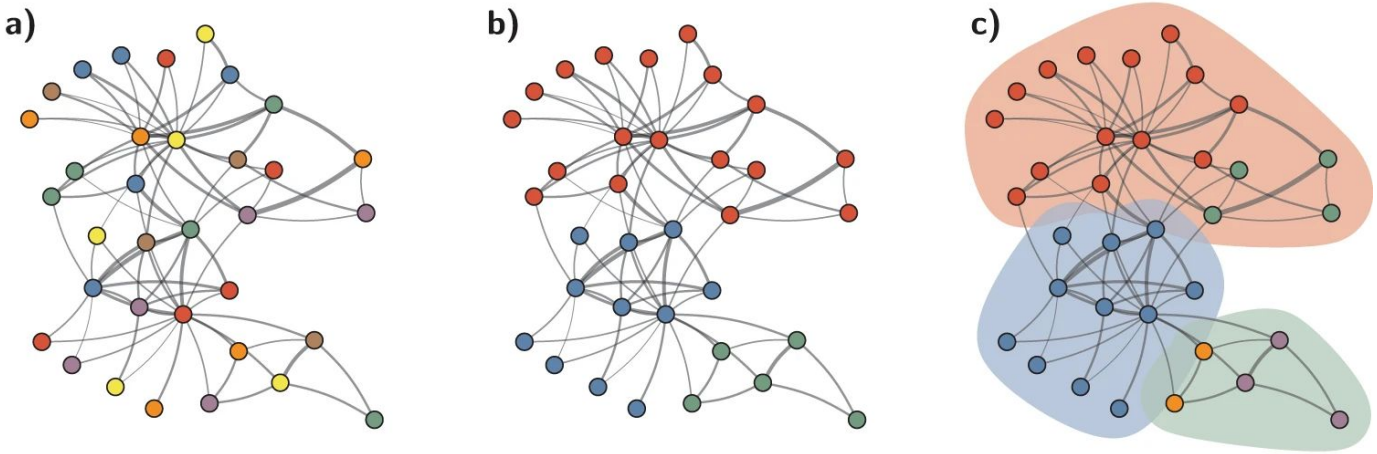
High-dimensional Data



High-dimension → low-dimension





Thrun and Habil, 2020

Clustering (Leiden)